

STABILITY OF FINITE-DIFFERENCE DISCRETIZATION ON CURVILINEAR MOVING OVERSET GRIDS*

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Abstract. Weak overset-grid coupling of summation-by-parts (SBP) finite-difference operators with simultaneous-approximation-term (SAT) penalties is a standard route to provably stable discretizations of hyperbolic initial-boundary-value problems on composite domains. Existing energy and eigenvalue proofs of stability for such schemes are stated for static grids, where the semi-discrete system matrix M is constant in time and stability reduces to a linear-algebraic (Lyapunov or eigenvalue) condition on M . Under grid motion, the interpolation weights and metric terms that couple the component grids vary with time, so the semi-discrete system is genuinely non-autonomous, $d\Phi/d\tau = M(\tau)\Phi + \mathbf{b}(\tau)$. We show, by an explicit model calculation, that verifying the eigenvalue or Lyapunov condition at each *frozen* time τ is necessary but not sufficient to conclude boundedness of $\Phi(\tau)$: the missing term is the energy-weight derivative $d\hat{H}/d\tau$, whose omission is not justified by any grid-refinement argument, since its point-wise $O(h)$ smallness is exactly compensated, in aggregate, by the $O(1/h)$ number of grid points it acts on. We then prove a rigorous, non-autonomous energy-stability theorem for the weak overset coupling of SBP-SAT operators under prescribed rigid-body grid translation, by (i) identifying the discrete geometric conservation law (GCL) that makes the metric-derivative contribution to $d\hat{H}/d\tau$ collapse to a single scalar “acceleration” term for translating grids, and (ii) closing the resulting differential energy inequality with Grönwall’s lemma under a mild, verifiable boundedness assumption on the interpolation operators. The resulting bound is explicit, computable from the prescribed grid-motion law, and is illustrated numerically for the sinusoidally translating overset configuration used throughout the paper. We further outline, without claiming a complete proof, the additional structure (entropy variables, entropy-conservative fluxes, and a discrete GCL compatible with the SBP operator) that extending this analysis to the nonlinear compressible Euler and Navier–Stokes equations would require.

Key words. energy stability, overset interface treatment, moving grids

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1. Introduction. Overset (Chimera) grids provide a flexible approach to flow simulation in domains containing bodies that move relative to one another. Each component can be represented by an independently generated structured grid, while interpolation transfers solution information across the regions of overlap. This arrangement accommodates relative motion without requiring the component grids to deform with the entire configuration, making it useful for applications such as store separation and fluid–structure interaction [13]. The resulting flexibility introduces a numerical challenge: as grids move, both the interpolation weights and the donor stencils can change, and the interface treatment must control the effect of these changes on the evolution of the discrete solution.

The development of overset methods has addressed geometric flexibility, interpolation accuracy, conservation, and stability through distinct constructions. Chesshire and Henshaw [4] described composite overlapping meshes for partial differential equations, including applications to compressible flow. Sherer and Scott [19] subsequently combined high-order compact finite differences with high-order interpolation on gen-

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eral overset grids, demonstrating the potential of this approach for viscous flows and acoustic propagation. Conservation across overlaps requires additional constraints on the transfer operators, as illustrated by the conservative interpolation scheme of Chesshire and Henshaw [3]. The stability of the coupled problem has motivated complementary developments: Bodony et al. [2] analyzed stable overset methods for computational aeroacoustics, and Reichert, Heath, and Bodony [14] developed energy-stable overlapping domain discretizations using generalized SBP operators. At the continuous level, Kopriva, Nordström, and Gassner [9] established well-posedness and conservation for penalized overset formulations under conditions on the coupling matrices, using an energy that accounts explicitly for the overlap. Together, these studies motivate treating the interfaces as part of the stability analysis of the complete coupled problem. For the translating-grid configuration studied here, this requires tracking the interface operators throughout the prescribed motion.

Summation-by-parts (SBP) finite-difference operators, combined with simultaneous-approximation-term (SAT) penalties for weak enforcement of boundary and interface conditions, provide a systematic framework for energy estimates for high-order discretizations of hyperbolic and hyperbolic-parabolic initial-boundary-value problems [8, 20]. For static overset grids, Sharan, Pantano, and Bodony [18] developed time-stable SBP-SAT closures in which interpolated data are imposed on incoming characteristics. Their analysis provides the starting point for the moving-grid formulation considered here. At a fixed spatial resolution, a constant system matrix with eigenvalues strictly in the open left half-plane admits a symmetric positive definite Lyapunov weight and yields exponential decay of the homogeneous solution [1, 15]. This autonomous result does not, however, directly establish stability when the grid geometry and interpolation operators vary with time.

Grid motion introduces a second requirement: the geometric quantities in the transformed equations must be treated consistently with the spatial and temporal discretizations. In an arbitrary Lagrangian–Eulerian formulation, the geometric conservation law (GCL) relates the evolution of the mapping Jacobian to the grid velocity; its discrete counterpart is central to preserving a uniform solution on a moving mesh. Energy and entropy analyses on moving meshes must also account for this geometric evolution. For example, Schnücke et al. [16] constructed moving-mesh discontinuous Galerkin spectral element schemes with the SBP property and established a discrete entropy balance using suitable volume and interface fluxes. These results demonstrate that moving-mesh stability can be obtained with compatible geometric and interface treatments. For overset grids, the transfer of data between overlapping component grids introduces additional coupling terms that must be included in the energy estimate.

Related moving-grid studies clarify the roles of geometric consistency and energy control. Visbal and Gaitonde [22] investigated high-order finite differences on curvilinear and deforming meshes, including the treatment of metric terms needed for free-stream preservation. For moving hexahedral meshes, Kopriva et al. [10] derived a skew-symmetric discontinuous Galerkin spectral element approximation with provable stability, conservation, and free-stream preservation for constant-coefficient problems. Yamaleev et al. [23] extended entropy-stable spectral collocation methods to the three-dimensional compressible Navier–Stokes equations on dynamic unstructured grids, enforcing discrete geometric conservation laws in the construction. More recently, Lundquist, Malan, and Nordström [12] developed an SBP-based method-of-lines framework for energy-stable arbitrary Lagrangian–Eulerian discretizations and distinguished the stability requirements from the additional property of exact free-

stream preservation. These results emphasize that geometric conservation and energy stability must be assessed through the particular discrete formulation. They also provide context for the present analysis: rigid translation simplifies the geometric dependence, but the resulting time-dependent energy weight and interpolation between overlapping grids still contribute to the energy balance and must be estimated explicitly.

Mapping each component grid to a fixed computational domain produces grid-velocity-dependent coefficients and time-dependent interpolation operators. The semi-discrete problem therefore takes the non-autonomous form

$$(1.1) \quad \frac{d\Phi}{d\tau} = M(\tau)\Phi + \mathbf{b}(\tau),$$

where Φ collects the component-grid solutions and $\mathbf{b}$ contains the prescribed boundary forcing. Frozen-time eigenvalues describe the matrix $M(\tau)$ at an individual instant, whereas stability depends on the evolution generated by the full matrix family. In general, having a Hurwitz matrix at every instant is not sufficient to ensure bounded solutions of a time-varying system [1, 15]. Similarly, a Lyapunov inequality verified separately at each instant is insufficient if its weight varies with time. Differentiating the weighted energy $E(\tau) = \Phi^T \hat{H}(\tau) \Phi$ introduces the additional term $\Phi^T (d\hat{H}/d\tau) \Phi$, which must be retained alongside the spatial and interpolation contributions. The scaling of the SBP quadrature weights alone does not justify neglecting this term: $O(h)$ contributions at $O(h^{-1})$ grid points can yield an $O(1)$ contribution to the energy rate.

1.1. Contributions and scope. This paper examines these effects for linear scalar advection on three overlapping one-dimensional grids, with fixed left and right grids and a rigidly translating middle grid. We assume that the relative advection speed remains positive, so that the direction of information transfer does not change. This model retains the time dependence of both the grid-velocity coefficients and the overset interpolation while allowing the moving-grid contribution to the energy rate to be expressed explicitly. The analysis yields an energy bound on finite time intervals for each fixed spatial discretization, under a uniform boundedness assumption on the interpolation weights. Because the estimated coupling rate depends on the smallest eigenvalue of the discrete norm, the bound is not uniform under grid refinement and does not by itself establish a convergence theorem or a fully discrete stability result.

The paper is organized around the following contributions:

- Section 2 formulates the moving overset SBP-SAT discretization and identifies the time-dependent coefficients and interpolation operators.
- Section 3 establishes the frozen-time spectral result and explains why it does not control the energy evolution of the non-autonomous system.
- Section 4 uses rigid translation to factor the moving-grid energy weight into a time-dependent scalar and a fixed matrix. Retaining its derivative and bounding the interpolation terms yields a Grönwall estimate with separate contributions from grid acceleration and overset coupling.
- Section 5 discusses the additional entropy and geometric compatibility requirements for an extension to the compressible Euler and Navier–Stokes equations; such an extension is outside the scope of the present proof.
- Section 6 reports accuracy and frozen-time spectra for sinusoidal grid motion and evaluates the grid-motion contribution to the theoretical growth-rate bound.

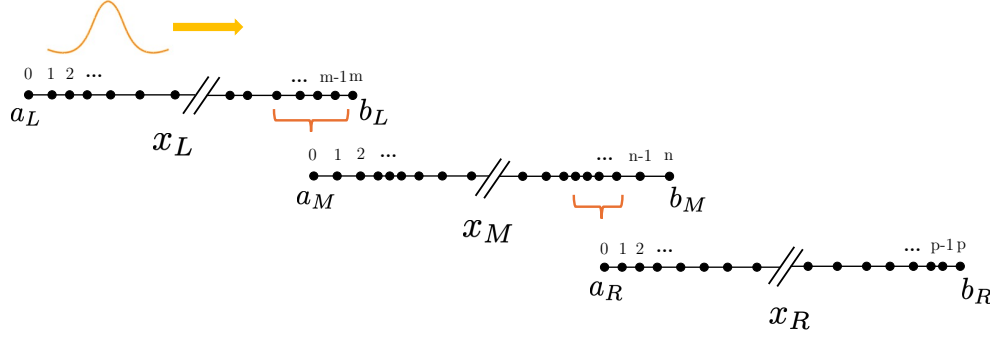


FIG. 1. Schematic of the one-dimensional overlapping grids on which Eq. (2.1) is solved.

- Section 7 examines the grid dependence of the bound and discusses possible routes to a uniform stability estimate.

2. Problem formulation: moving overset method for scalar advection equation. Consider the one-dimensional linear advection equation

$$(2.1) \quad \frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0, \quad -1 \leq x \leq 1, \quad t \geq 0,$$

on three overlapping (overset) grids: a left grid $-1 = a_L \leq x \leq b_L$, a middle grid $a_M(t) \leq x \leq b_M(t)$, and a right grid $a_R \leq x \leq b_R = 1$, as shown in Fig. 1. The left and right grids are static, whereas the middle grid moves according to

$$(2.2) \quad \frac{dx_M}{dt} = \beta(t),$$

so that its left and right boundaries remain within the neighboring grids. Assuming $c > 0$, the initial and boundary conditions are, respectively,

$$(2.3) \quad u(x, 0) = f(x), \quad u(-1, t) = g(t).$$

To accommodate moving and spatially stretched grids, we introduce the general curvilinear coordinate transformation

$$(2.4) \quad \tau = t, \quad \xi = \xi(x, t).$$

Applying the chain rule, the Cartesian derivatives can be expressed in terms of the curvilinear derivatives using

$$(2.5) \quad \frac{\partial}{\partial t} = \frac{\partial}{\partial \tau} + \xi_t \frac{\partial}{\partial \xi}, \quad \frac{\partial}{\partial x} = \xi_x \frac{\partial}{\partial \xi},$$

where the shorthand notation $\xi_t = \partial \xi / \partial t$ and $\xi_x = \partial \xi / \partial x$ is used. Similarly, the curvilinear derivatives can be expressed in terms of the Cartesian derivatives using

$$(2.6) \quad \frac{\partial}{\partial \tau} = \frac{\partial}{\partial t} + x_\tau \frac{\partial}{\partial x}, \quad \frac{\partial}{\partial \xi} = x_\xi \frac{\partial}{\partial x}.$$

Equations (2.5) and (2.6) can be written as, respectively,

$$(2.7) \quad \begin{bmatrix} \partial / \partial t \\ \partial / \partial x \end{bmatrix} = \begin{bmatrix} 1 & \xi_t \\ 0 & \xi_x \end{bmatrix} \begin{bmatrix} \partial / \partial \tau \\ \partial / \partial \xi \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} \partial / \partial \tau \\ \partial / \partial \xi \end{bmatrix} = \begin{bmatrix} 1 & x_\tau \\ 0 & x_\xi \end{bmatrix} \begin{bmatrix} \partial / \partial t \\ \partial / \partial x \end{bmatrix}.$$

163 Inverting the second equation in (2.7) and comparing it with the first equation provides
 164 the metric relations

$$165 \quad (2.8) \quad \xi_t = -x_\tau \xi_x, \quad \xi_x = \frac{1}{x_\xi}.$$

166 Substituting (2.5) into (2.1) provides

$$167 \quad (2.9) \quad \frac{\partial u}{\partial \tau} + (\xi_t + c \xi_x) \frac{\partial u}{\partial \xi} = 0,$$

168 and using (2.8) in (2.9) yields

$$169 \quad (2.10) \quad \frac{\partial u}{\partial \tau} + (-x_\tau + c) \xi_x \frac{\partial u}{\partial \xi} = 0.$$

170 Here, x_τ incorporates the grid motion. For the static left and right grids, $x_\tau \equiv 0$;
 171 for the middle grid, $x_\tau = \beta(t)$ is, by (2.2), the same at every point of the grid (rigid
 172 translation carries no spatial stretching), and we assume

$$173 \quad (2.11) \quad -x_\tau + c > 0 \quad \text{on every grid, at every } \tau,$$

174 i.e., the grid speed never exceeds the wave speed c ; this fixes the upwind direction of
 175 the weak coupling.

176 Assuming $c > 0$ and $-x_\tau + c > 0$, a semi-discretization of (2.10) with the boundary
 177 condition (2.3) using a weak boundary/interface treatment on the three overset grids
 178 shown in Fig. 1 can be written as

$$179 \quad (2.12) \quad \frac{d\mathbf{u}}{d\tau} = -\mathcal{X}_L D_L \mathbf{u} - \tau_L \mathcal{X}_L H_L^{-1} \mathbf{e}_0^L (u_0 - g(\tau)),$$

$$180 \quad (2.13) \quad \frac{d\mathbf{v}}{d\tau} = -\mathcal{X}_M(\tau) D_M \mathbf{v} - \tau_M \mathcal{X}_M(\tau) H_M^{-1} \mathbf{e}_0^M (v_0 - \mathbf{I}_L(\tau)^T \mathbf{u}),$$

$$181 \quad (2.14) \quad \frac{d\mathbf{w}}{d\tau} = -\mathcal{X}_R D_R \mathbf{w} - \tau_R \mathcal{X}_R H_R^{-1} \mathbf{e}_0^R (w_0 - \mathbf{I}_M(\tau)^T \mathbf{v}),$$

183 where

$$184 \quad \mathbf{u} = [u_0 \quad \cdots \quad u_m]^T, \quad \mathbf{v} = [v_0 \quad \cdots \quad v_n]^T, \quad \mathbf{w} = [w_0 \quad \cdots \quad w_p]^T$$

185 are the solution vectors on the left, middle, and right grids, respectively. In (2.12)–
 186 (2.14), $\mathcal{X}_\kappa$, $\kappa \in \{L, M, R\}$, are diagonal matrices whose i -th diagonal entry is an
 187 approximation to $|-x_\tau + c| \xi_x$ at the i -th grid point of the respective grids. D_κ
 188 denote summation-by-parts first-derivative approximations of the form $D_\kappa = H_\kappa^{-1} Q_\kappa$
 189 [8, Appendix B.1], where H_κ is a diagonal, symmetric positive definite (s.p.d.) norm
 190 matrix and

$$191 \quad (2.15) \quad Q_\kappa + Q_\kappa^T = \text{diag}(-1, 0, \dots, 0, 1).$$

192 τ_κ denotes the penalty parameter, the vector $\mathbf{e}_0^\kappa = [1 \ 0 \ \cdots \ 0]^T$ selects the left
 193 boundary node, and $\mathbf{I}_L(\tau)$ contains the interpolation weights for the solution interpo-
 194 lation from the left grid to the first node of the middle grid. Similarly, $\mathbf{I}_M(\tau)$ enables
 195 interpolation from the middle grid to the first node of the right grid. Since the middle
 196 grid translates relative to the fixed left and right grids, the donor stencil and the

interpolation weights at each interface change with τ , so the interpolation operators are written as functions of τ

The discretization (2.13), for the middle grid, assumes $-x_\tau + c > 0$; therefore, the interpolated data is (weakly) imposed at the left boundary of the grid. The stability proof for $c < 0$ and $-x_\tau + c < 0$ (where the boundary condition must be imposed at the right boundary) will follow a similar procedure, as discussed below. The constraint $-x_\tau + c > 0$ for $c > 0$ (and, similarly, $-x_\tau + c < 0$ for $c < 0$) can be interpreted as the assumption that the moving grid velocity (typically associated with structure/body's motion) does not exceed the freestream velocity.

The scheme (2.12)–(2.14) can be collectively written as

$$(2.16) \quad \frac{d\Phi}{d\tau} = M(\tau)\Phi + \mathbf{b}(\tau), \quad \Phi = \begin{bmatrix} \mathbf{u} \\ \mathbf{v} \\ \mathbf{w} \end{bmatrix}.$$

where the matrix M , called the system or discretization matrix, is given by

$$(2.17) \quad M(\tau) = \begin{bmatrix} -\mathcal{X}_L H_L^{-1} Q_L - \tau_L \mathcal{X}_L H_L^{-1} E_0^L & 0 & 0 \\ \tau_M \mathcal{X}_M(\tau) H_M^{-1} \mathbf{e}_0^M \mathbf{I}_L(\tau)^T & -\mathcal{X}_M(\tau) H_M^{-1} Q_M - \tau_M \mathcal{X}_M(\tau) H_M^{-1} E_0^M & 0 \\ 0 & \tau_R \mathcal{X}_R H_R^{-1} \mathbf{e}_0^R \mathbf{I}_M(\tau)^T & -\mathcal{X}_R H_R^{-1} Q_R - \tau_R \mathcal{X}_R H_R^{-1} E_0^R \end{bmatrix},$$

and

$$(2.18) \quad E_0^\kappa = \mathbf{e}_0^\kappa (\mathbf{e}_0^\kappa)^T, \quad \mathbf{b}(\tau) = \begin{bmatrix} \tau_L \mathcal{X}_L H_L^{-1} \mathbf{e}_0^L g(\tau) \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}.$$

$M(\tau)$ is block lower triangular, where the sources of τ -dependence are $\mathcal{X}_M(\tau)$, because of the translating middle grid's metric term, and the interpolation operators $\mathbf{I}_L(\tau)$ and $\mathbf{I}_M(\tau)$, because of the continuously changing relative offset between the translating middle grid and its fixed neighbors.

3. Frozen-time stability analysis. Since $M(\tau)$ is block lower triangular, its eigenvalues at any fixed τ coincide with those of its block-diagonal part

$$(3.1) \quad \hat{M}(\tau) = \begin{bmatrix} -\mathcal{X}_L H_L^{-1} Q_L - \tau_L \mathcal{X}_L H_L^{-1} E_0^L & 0 & 0 \\ 0 & -\mathcal{X}_M(\tau) H_M^{-1} Q_M - \tau_M \mathcal{X}_M(\tau) H_M^{-1} E_0^M & 0 \\ 0 & 0 & -\mathcal{X}_R H_R^{-1} Q_R - \tau_R \mathcal{X}_R H_R^{-1} E_0^R \end{bmatrix},$$

which is independent of the interpolation operators $\mathbf{I}_L$ and $\mathbf{I}_M$. $\mathcal{X}_\kappa$ is an approximation to $|-x_\tau + c| \xi_x$, which is positive definite. We define a s.p.d. matrix

$$(3.2) \quad \hat{H} = \begin{bmatrix} H_L \mathcal{X}_L^{-1} & & \\ & H_M \mathcal{X}_M^{-1} & \\ & & H_R \mathcal{X}_R^{-1} \end{bmatrix}.$$

LEMMA 3.1 (Frozen-time dissipativity). *For every fixed τ ,*

$$(3.3) \quad \hat{H} \hat{M} + \hat{M}^T \hat{H} = \begin{bmatrix} -(Q_L + Q_L^T) - 2\tau_L E_0^L & 0 & 0 \\ 0 & -(Q_M + Q_M^T) - 2\tau_M E_0^M & 0 \\ 0 & 0 & -(Q_R + Q_R^T) - 2\tau_R E_0^R \end{bmatrix},$$

225 where

$$226 \quad (3.4) \quad -(Q_\kappa + Q_\kappa^T) - 2\tau_\kappa E_0^\kappa = \begin{bmatrix} 1 - 2\tau_\kappa & & & & \\ & 0 & & & \\ & & \ddots & & \\ & & & 0 & \\ & & & & -1 \end{bmatrix}$$

227 for $\kappa \in \{L, M, R\}$. The matrix in (3.4) is negative semi-definite for $\tau_\kappa \geq \frac{1}{2}$, and every
 228 eigenvalue of $\hat{M}(\tau)$ — equivalently, of $M(\tau)$ — has non-positive real part. Moreover,
 229 for $\tau_\kappa > \frac{1}{2}$, no eigenvalue has zero real part, so every eigenvalue of $M(\tau)$ lies strictly
 230 in the open left half-plane, for every fixed τ .

231 *Proof.* Using (2.15), each diagonal block of (3.3) reduces algebraically to (3.4),
 232 which is negative semi-definite for $\tau_\kappa \geq \frac{1}{2}$.

233 Let $\mathbf{s}_i^\kappa = [s_{i,0}^\kappa \ \cdots \ s_{i,N}^\kappa]^T$, where $N = m, n, p$ are the number of grid points
 234 (see Fig. 1) for $\kappa = L, M, R$, respectively, be the eigenvector corresponding to the
 235 eigenvalue λ_i^κ of $\hat{M}_\kappa = -\mathcal{X}_\kappa H_\kappa^{-1} Q_\kappa - \tau_\kappa \mathcal{X}_\kappa H_\kappa^{-1} E_0^\kappa$, then Eq. (3.4) provides

$$236 \quad (3.5) \quad (\mathbf{s}_i^\kappa)^* (\hat{H}_\kappa \hat{M}_\kappa + \hat{M}_\kappa^* \hat{H}_\kappa) \mathbf{s}_i^\kappa = (1 - 2\tau_\kappa) |s_{i,0}^\kappa|^2 - |s_{i,N}^\kappa|^2,$$

237 where $*$ denotes the conjugate transpose. Furthermore, using $\hat{M}_\kappa \mathbf{s}_i^\kappa = \lambda_i^\kappa \mathbf{s}_i^\kappa$,

$$238 \quad (3.6) \quad (\mathbf{s}_i^\kappa)^* (\hat{H}_\kappa \hat{M}_\kappa + \hat{M}_\kappa^* \hat{H}_\kappa) \mathbf{s}_i^\kappa = (\mathbf{s}_i^\kappa)^* \hat{H}_\kappa (\hat{M}_\kappa \mathbf{s}_i^\kappa) + (\hat{M}_\kappa \mathbf{s}_i^\kappa)^* \hat{H}_\kappa \mathbf{s}_i^\kappa \\ 239 \quad = [\lambda_i^\kappa + (\lambda_i^\kappa)^*] (\mathbf{s}_i^\kappa)^* \hat{H}_\kappa \mathbf{s}_i^\kappa.$$

242 Eqs. (3.5) and (3.6) provide

$$243 \quad (3.7) \quad \operatorname{Re}(\lambda_i^\kappa) = \frac{(1 - 2\tau_\kappa) |s_{i,0}^\kappa|^2 - |s_{i,N}^\kappa|^2}{2 (\mathbf{s}_i^\kappa)^* \hat{H}_\kappa \mathbf{s}_i^\kappa}.$$

244 Since $\hat{H}_\kappa$ is positive definite, $\tau_\kappa \geq \frac{1}{2}$ in (3.7) implies $\operatorname{Re}(\lambda_i^\kappa) \leq 0$. For $\tau_\kappa > \frac{1}{2}$,
 245 $\operatorname{Re}(\lambda_i^\kappa) = 0$ if and only if $s_{i,0}^\kappa = s_{i,N}^\kappa = 0$. The reduced row echelon form of $\hat{M}_\kappa(\tau) - \lambda I$
 246 with its first and last columns removed has full rank (see, e.g., [17, Lemmas 1.3–1.4]),
 247 so $s_{i,0}^\kappa = s_{i,N}^\kappa = 0$ forces $\mathbf{s}_i^\kappa = \mathbf{0}$, a contradiction. Hence, $\operatorname{Re}(\lambda_i^\kappa) < 0$ for every
 248 eigenvalue, at every fixed τ . $\square$

249 Lemma 3.1 is the exact analogue, at each frozen instant, of the static-grid stability
 250 result; its proof is unchanged because τ enters $\hat{M}(\tau)$ and $\hat{H}(\tau)$ only through the
 251 positive scalar-like factor $\mathcal{X}_M(\tau)$, which cancels identically in the Lyapunov product
 252 (3.3) irrespective of its value. Notably, the proof is independent of $\mathbf{I}_L$ and $\mathbf{I}_M$: the
 253 block-triangular structure of $M(\tau)$ means the spectrum, at each frozen τ , does not
 254 depend on the interpolation operators.

255 **Remark 3.2 (Frozen-time dissipativity does not imply stability).** Lemma 3.1
 256 establishes that $M(\tau)$ is Hurwitz for every fixed τ . For the static case (where M is
 257 constant), this implies the boundedness of solutions of $d\Phi/d\tau = M\Phi$. For the time-
 258 dependent system (2.16), it is not: the solution operator (state-transition matrix) of
 259 a linear time-varying system is, in general, *not* $\exp(\int_0^\tau M(s) ds)$, and pointwise Hur-
 260 witz spectra do not control its norm. This is a classical observation in linear systems
 261 theory: a family $M(\tau)$ can be Hurwitz, with eigenvalues bounded uniformly away

from the imaginary axis at every instant, while the corresponding time-dependent system has solutions that grow without bound, if the eigenvectors of $M(\tau)$ rotate on a time scale comparable to or faster than the decay rate that the eigenvalues would suggest [1, 15]. The correct route to a stability statement for (2.16) is therefore the direct, time-dependent energy method, *i.e.*, differentiate the time-dependent weighted energy $E(\tau) = \Phi^T \hat{H}(\tau) \Phi$ along solutions of (2.16),

$$(3.8) \quad \frac{dE}{d\tau} = \Phi^T \left[\hat{H}(\tau) M(\tau) + M(\tau)^T \hat{H}(\tau) + \frac{d\hat{H}}{d\tau} \right] \Phi + 2\Phi^T \hat{H}(\tau) b(\tau),$$

which contains a term, $\Phi^T (d\hat{H}/d\tau) \Phi$, absent from the frozen-time Lyapunov identity (3.3).

4. Time-dependent stability analysis. For pure rigid translation, $x_\tau = \beta(\tau)$ is spatially uniform on the middle grid and ξ_x is independent of τ (the mapped shape of the grid in computational space does not change, only its physical location does); this is the discrete statement of the geometric conservation law for a translating grid. Consequently

$$(4.1) \quad \mathcal{X}_M(\tau) = a(\tau) \Xi_x, \quad a(\tau) := c - \beta(\tau) > 0,$$

with $\Xi_x = \text{diag}(\xi_{x,0}, \dots, \xi_{x,n})$ a *fixed* (time-independent), s.p.d. diagonal matrix, and $a(\tau)$ a strictly positive scalar by (2.11). $\xi_{x,i}$ for $0 \leq i \leq n$ denotes the value of ξ_x at the i -th grid point on the middle grid. Substituting (4.1) into the middle-grid block of (3.1) and (3.2) gives

$$(4.2) \quad \hat{M}_M(\tau) = a(\tau) \hat{M}_M^{\text{ref}}, \quad \hat{H}_M(\tau) = \frac{1}{a(\tau)} \hat{H}_M^{\text{ref}},$$

with $\hat{M}_M^{\text{ref}} = -\Xi_x H_M^{-1} Q_M - \tau_M \Xi_x H_M^{-1} E_0^M$ and $\hat{H}_M^{\text{ref}} = H_M \Xi_x^{-1}$ both fixed. For rigid translation, the entire τ -dependence of the moving block collapses into a single positive scalar multiplying a fixed operator. Two immediate consequences follow. First, the frozen-time Lyapunov product is τ -independent for this block,

$$(4.3) \quad \hat{H}_M(\tau) \hat{M}_M(\tau) + \hat{M}_M(\tau)^T \hat{H}_M(\tau) = \hat{H}_M^{\text{ref}} \hat{M}_M^{\text{ref}} + (\hat{M}_M^{\text{ref}})^T \hat{H}_M^{\text{ref}} =: N_M^{\text{ref}} \preceq 0,$$

the scalar $a(\tau)$ having canceled identically. Second,

$$(4.4) \quad \frac{d\hat{H}_M}{d\tau} = -\frac{\dot{a}(\tau)}{a(\tau)^2} \hat{H}_M^{\text{ref}} = -\frac{\dot{a}(\tau)}{a(\tau)} \hat{H}_M(\tau),$$

and $\hat{H}_L, \hat{H}_R$ are time-independent, so $d\hat{H}/d\tau = \text{blkdiag}(0, d\hat{H}_M/d\tau, 0)$. Equations (4.3)–(4.4) identify, respectively, the purely dissipative part of (3.8) and the metric-derivative correction that, as shown by (4.4), is controlled by the single, explicit, dimensionless quantity $\dot{a}/a$, *i.e.*, by the grid acceleration relative to the instantaneous margin $a = c - \beta$ between grid speed and wave speed.

4.1. Interpolation boundedness. The off-diagonal (coupling) part of $\Phi^T [\hat{H}_M + M^T \hat{H}] \Phi$, computed from (3.2) and (2.17) provides

$$(4.5) \quad C(\tau) = 2\tau_M v_0 (\mathbf{I}_L(\tau)^T \mathbf{u}) + 2\tau_R w_0 (\mathbf{I}_M(\tau)^T \mathbf{v}),$$

using $\hat{H}_M(\tau) \cdot \tau_M \mathcal{X}_M(\tau) H_M^{-1} \mathbf{e}_0^M \mathbf{I}_L(\tau)^T = \tau_M \mathbf{e}_0^M \mathbf{I}_L(\tau)^T$ (the $\mathcal{X}_M, H_M$ factors cancel exactly, independent of (4.1), at each frozen τ). Unlike (4.3), this term is not sign-definite; it must be bounded rather than discarded.

Assumption 4.1 (Bounded interpolation). The interpolation operators satisfy, for every τ , $|\mathbf{I}_L(\tau)^T \mathbf{u}| \leq \Lambda_L \|\mathbf{u}\|_2$ and $|\mathbf{I}_M(\tau)^T \mathbf{v}| \leq \Lambda_M \|\mathbf{v}\|_2$ for constants Λ_L, Λ_M independent of the grid spacing and of τ , *i.e.*, uniform over the whole family of interpolation operators the moving grid realizes, not just their value at a single instant.

Assumption 4.1 is mild and directly verifiable for the interpolation stencils commonly used in overset schemes: it states that the interpolation weights, viewed as a row vector, have bounded Euclidean norm, *i.e.*, a bounded (grid-independent) Lebesgue-type constant. For r -point Lagrange interpolation with barycentric weights $\{\ell_i\}_{i=1}^r$ this holds with $\Lambda = (\sum_i \ell_i^2)^{1/2}$; for linear interpolation ($r = 2$), $\Lambda \leq 1$ exactly (the weights lie on the unit interval and sum to one), and for the cubic ($r = 4$) stencils used with high-order schemes, Λ is likewise a bounded, grid-independent constant determined solely by the normalized stencil offsets and interpolation location. For cubic interpolation with the standard four-point stencil and the target between the two central nodes, used in this study, $\Lambda \leq 1$, as in the linear case (see Remark 6.1).

THEOREM 4.2 (Time-dependent energy stability under rigid translation). *Let $\tau_L = \tau_M = \tau_R = 1$, let $g \equiv 0$ in (2.3), and let Assumption 4.1 hold. Then the energy $E(\tau) = \Phi(\tau)^T \hat{H}(\tau) \Phi(\tau)$ of the solution of (2.16) satisfies*

$$(4.6) \quad E(\tau) \leq E(0) \exp\left(\int_0^\tau [\rho(s) + \gamma(s)] ds\right), \quad \gamma(s) = \left|\frac{\dot{a}(s)}{a(s)}\right| = \frac{|\ddot{x}_M(s)|}{c - \dot{x}_M(s)},$$

where $\rho(\tau) = (\Lambda_L^2 + \Lambda_M^2)/\lambda_{\min}(\hat{H}(\tau))$ and $\lambda_{\min}(\hat{H}(\tau))$ is the smallest eigenvalue of $\hat{H}(\tau)$. In particular, for any prescribed C^2 grid motion, given by $x_M(\tau)$, satisfying (2.11) on a bounded time interval $[0, T]$, $E(\tau)$ remains finite for all $\tau \in [0, T]$: the scheme does not blow up under the moving-grid coupling.

Proof. With $\tau_\kappa = 1$, the diagonal (dissipative) part of $\hat{H}M + M^T \hat{H}$ from (3.3) is $\text{blkdiag}(A_L, A_M, A_R)$ with $A_\kappa = \text{diag}(-1, 0, \dots, 0, -1) \preceq 0$, and by (4.5) and Young's inequality,

$$\begin{aligned} C(\tau) &= 2v_0(\mathbf{I}_L^T \mathbf{u}) + 2w_0(\mathbf{I}_M^T \mathbf{v}) \leq v_0^2 + (\mathbf{I}_L^T \mathbf{u})^2 + w_0^2 + (\mathbf{I}_M^T \mathbf{v})^2 \\ &\leq v_0^2 + w_0^2 + \Lambda_L^2 \|\mathbf{u}\|_2^2 + \Lambda_M^2 \|\mathbf{v}\|_2^2. \end{aligned}$$

Adding the diagonal part (whose v_0^2, w_0^2 contributions from A_M, A_R at $\tau_M = \tau_R = 1$ are $-v_0^2, -w_0^2$, exactly canceling the terms just isolated) and dropping the remaining non-positive boundary terms $-u_0^2 - u_m^2 - v_n^2 - w_p^2$ gives

$$\begin{aligned} (4.7) \quad \Phi^T [\hat{H}(\tau)M(\tau) + M(\tau)^T \hat{H}(\tau)] \Phi &\leq \Lambda_L^2 \|\mathbf{u}\|_2^2 + \Lambda_M^2 \|\mathbf{v}\|_2^2 \\ &\leq (\Lambda_L^2 + \Lambda_M^2) \|\Phi\|_2^2 \leq \rho(\tau) E(\tau), \end{aligned}$$

using $\|\Phi\|_2^2 \leq E(\tau)/\lambda_{\min}(\hat{H}(\tau))$. By (4.4) and $\hat{H}_M(\tau) \preceq \hat{H}(\tau)$ (block inclusion),

$$(4.8) \quad \Phi^T \left(\frac{d\hat{H}}{d\tau} \right) \Phi = -\frac{\dot{a}(\tau)}{a(\tau)} \mathbf{v}^T \hat{H}_M(\tau) \mathbf{v} \leq \gamma(\tau) E(\tau).$$

With $g \equiv 0$, $\mathbf{b} \equiv \mathbf{0}$, so (3.8), (4.7), and (4.8) give the scalar differential inequality $\dot{E}(\tau) \leq [\rho(\tau) + \gamma(\tau)]E(\tau)$, and Grönwall's lemma yields (4.6). Finiteness of the bound on $[0, T]$ follows because $x_M \in C^2([0, T])$, ρ is continuous and $a(\tau) \geq c - \max |\dot{x}_M| > 0$ is bounded away from zero by (2.11), so $\rho, \gamma \in L^1(0, T)$. $\square$

Remark 4.3 (Interpretation, and comparison with the frozen-time result). Theorem 4.2 closes exactly the gap left open by Lemma 3.1: it controls the true, non-autonomous energy $\Phi(\tau)^T \hat{H}(\tau) \Phi(\tau)$, rather than the spectrum of the frozen operator, and it does so by an explicit, a priori computable bound in terms of the prescribed grid-motion law alone. The rate $\gamma(\tau)$ is precisely the term that Section 3 showed the frozen-time analysis silently discards; Theorem 4.2 shows it is controllable, not negligible. The rate $\rho(\tau)$ is a genuinely different contribution, tracing to the multi-grid coupling rather than to grid motion; it does not vanish even for a static overset configuration ($\gamma \equiv 0$), and it scales as $\lambda_{\min}(\hat{H})^{-1}$, which is grid-spacing dependent (the SBP norm H_κ scales with h_κ). This limits the strength of the estimate: for each *fixed* discretization, ρ is a fixed finite constant, so the scheme is stable at that resolution over any bounded time horizon; the bound is simply not, and is not claimed to be, uniform in the grid spacing, matching the corresponding caveat already present in the static-grid theory [18, Remark following Thm. 1]. Section 7 discusses the distinction between fixed-resolution boundedness and stability under refinement, and possible approaches to obtaining a uniform bound.

Remark 4.4 (Sharpness). The bound (4.6) is an a priori, worst-case estimate obtained from Young’s inequality and Grönwall’s lemma, and neither step is expected to be tight along the actual dynamics. The Young step in the proof, $2v_0(\mathbf{I}_L(\tau)^T u) \leq v_0^2 + (\mathbf{I}_L(\tau)^T u)^2$, is an equality only when $v_0 = \mathbf{I}_L(\tau)^T u$ pointwise in τ (and likewise for $w_0 = \mathbf{I}_M(\tau)^T v$); the subsequent Cauchy–Schwarz step $(\mathbf{I}_L(\tau)^T u)^2 \leq \Lambda_L^2 \|\mathbf{u}\|_2^2$ of Assumption 4.1 is a separate equality condition, requiring $\mathbf{u}$ itself to be aligned with the interpolation weights. There is no reason for the true solution to satisfy either condition, let alone both simultaneously, so the bound is not expected to be quantitatively sharp. Section 6.2 evaluates it explicitly for the grid motion used in this paper’s numerical experiments and compares it with the actually observed behavior.

4.2. Beyond rigid translation.

Remark 4.5 (General rigid-body and deforming motion). The scalar-factorization (4.1)–(4.2) is special to rigid translation: it is what makes $\mathcal{X}_M \mathcal{X}_M^{-1}$ collapse to the single scalar $-\dot{a}/a$, with no residual matrix structure to control. For a rotating or deforming grid, x_τ is no longer spatially uniform and ξ_x is no longer time-independent, so $\mathcal{X}_M(\tau)$ is not a scalar multiple of a fixed matrix, and the continuous GCL identity $\partial_\tau \xi_x = \partial_\xi(x_\tau \xi_x)$ has a non-vanishing right-hand side. The natural extension of Theorem 4.2 to this setting is to require the discrete metric time-derivative entering $\mathcal{X}_M(\tau)$ to be computed with the *same* SBP operator D_M used for spatial derivatives, i.e., writing $\xi_x(\tau) := \text{diag}(\Xi_x(\tau)) \in \mathbb{R}^{n+1}$ and $\mathbf{x}_\tau(\tau)$ for the (now nodal, not scalar) vector of grid velocities on the middle grid, to enforce the discrete GCL, as an identity between vectors of nodal values,

$$(4.9) \quad \frac{d\xi_x}{d\tau}(\tau) \stackrel{!}{=} D_M(\mathbf{x}_\tau(\tau) \odot \xi_x(\tau)),$$

where $\odot$ denotes the entrywise (Hadamard) product, rather than evaluating $\partial_\tau \xi_x$ analytically and independently of D_M . When (4.9) holds exactly, the SBP telescoping property (2.15) converts the right-hand side of (4.9), once inserted into $\Phi^T(d\hat{H}_M/d\tau)\Phi$, into a boundary term of the same algebraic form as the dissipative boundary terms already appearing in (3.3), plus a remaining “acceleration” term analogous to $\gamma(\tau)E(\tau)$ in Theorem 4.2. We regard the construction of this general, GCL-compatible non-autonomous stability theorem, and its extension to rotating and

deforming multi-dimensional overset grids, as an important direction for future work, and do not claim it here.

5. Outlook: extension to nonlinear systems. The energy method of Sections 3–4 is intrinsically linear: it uses the fixed s.p.d. weight $\hat{H}(\tau)$ to define the energy norm, and linearity of (2.16) to write $d(\Phi^T \hat{H} \Phi)/d\tau$ as a quadratic form in Φ . Neither step is available for the nonlinear compressible Euler or Navier–Stokes equations. We record here, without proof, the additional structure that a corresponding non-autonomous stability result for the nonlinear equations would require, so as to scope it honestly as future work rather than to claim it as a consequence of Theorem 4.2.

1. *Entropy variables in place of a fixed energy weight.* The role of $\hat{H}$ is played, for a nonlinear hyperbolic system with a convex entropy $\eta(\mathbf{q})$, by the entropy/conservative-variable change of variables $\mathbf{w} = \partial\eta/\partial\mathbf{q}$, which symmetrizes the system and furnishes a state-dependent (rather than merely time-dependent) analogue of the energy norm [21].
2. *Entropy-conservative interior fluxes and entropy-stable SATs.* Discrete entropy stability on each component grid requires replacing the SBP first-derivative operator acting on the flux by a two-point entropy-conservative numerical flux satisfying a discrete chain rule compatible with $Q_\kappa + Q_\kappa^T$ [5], and replacing the linear characteristic penalty (2.13)–(2.14) by an entropy-stable SAT formulated in the entropy variables at the overset interface.
3. *A discrete GCL for the nonlinear metric Jacobian.* Section 4.2’s discrete GCL (4.9) must be enforced for the full, multi-dimensional, generally rotating or deforming metric Jacobian entering the contravariant velocities of the curvilinear flux divergence, using the same SBP operators used for the flux divergence, so that its contribution to the entropy-weighted energy identity again telescopes to boundary terms via summation by parts.
4. *Closure by Grönwall.* Given the above, the remaining “acceleration” contribution — the nonlinear analogue of $\gamma(\tau)$ in Theorem 4.2 — would need to be bounded in terms of the prescribed grid-motion law and closed by Grönwall’s inequality exactly as in Section 4, now applied to the nonlinear entropy functional rather than to the quadratic energy $\Phi^T \hat{H} \Phi$.

Each ingredient above is individually well developed in the literature; assembling them into a non-autonomous stability proof for moving-overset entropy-stable SBP-SAT discretizations of the compressible Navier–Stokes equations is, to our knowledge, open, and we leave it for future work rather than attempt it within the scope of the present, deliberately narrow, model-problem analysis.

6. Numerical verification. The scalar advection equation (2.10) is solved on the overset configuration of Section 2 with static left and right grids $(a_L, b_L) = (-1, -0.1)$, $(a_R, b_R) = (0.1, 1)$, and a sinusoidally translating middle grid

$$(6.1) \quad x_M(t) = A \sin(\omega t), \quad A = 0.1, \quad \omega = 2\pi/T, \quad T = 1,$$

so that $(a_M(t), b_M(t)) = (-0.35 + x_M(t), 0.35 + x_M(t))$, $c = 1$, and $\max_t |\dot{x}_M(t)| = A\omega \approx 0.628 < c$, consistent with (2.11). The initial/boundary data are set from the exact solution $u(x, t) = e^{-\beta(x-x_0-ct)^2}$ with $\beta = 80$, $x_0 = -0.4$. Three SBP-SAT schemes are used, denoted p -2 p - p by their interior order of accuracy on the left/middle/right grids ($p = 2, 3, 4$); linear (2-point) Lagrange interpolation is used with the 1-2-1 scheme and cubic (4-point) Lagrange interpolation with the 2-4-2 and 3-6-3 schemes.

6.1. Accuracy and frozen-time spectra. Figure 2 shows the solution at three times as it crosses the moving overset interfaces, together with a grid-refinement study confirming the theoretical global order of accuracy $p + 1$ [7]. Figure 3 shows the spectrum of $M(\tau)$, frozen at $\tau = 0.5$, for each scheme: every eigenvalue lies strictly in the open left half-plane, as guaranteed by Lemma 3.1 for every fixed τ and every scheme tested.

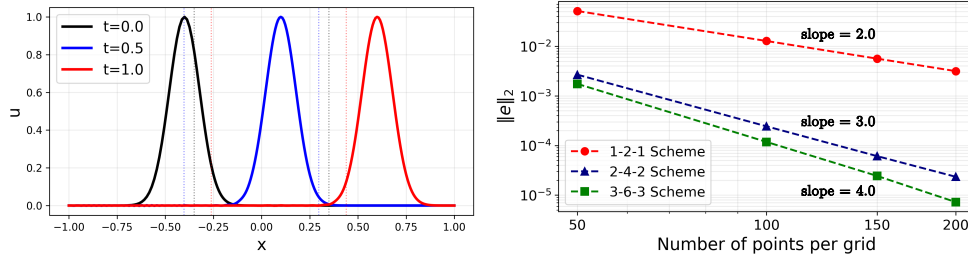


FIG. 2. (a) Solution from the 3-6-3 scheme at three times; vertical dashed lines mark the moving overset interface. (b) Global L_2 -error convergence, confirming the designed $p + 1$ order of accuracy.

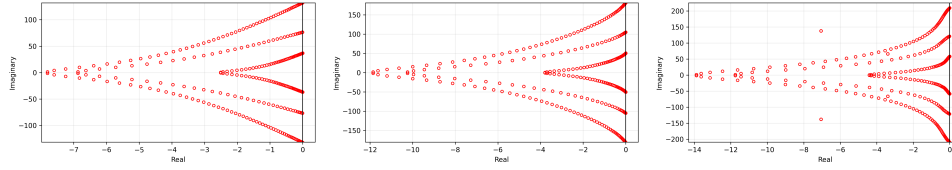


FIG. 3. Eigenvalue spectra of $M(\tau)$ at $\tau = 0.5$ for the (a) 1-2-1, (b) 2-4-2, and (c) 3-6-3 schemes, illustrating Lemma 3.1. These are pointwise-in-time spectra; they are necessary, but by Section 3, not by themselves sufficient, evidence for stability of the full non-autonomous system.

6.2. The Grönwall growth-rate bound. For the motion (6.1), $\dot{x}_M(\tau) = A\omega \cos(\omega\tau)$ and $\ddot{x}_M(\tau) = -A\omega^2 \sin(\omega\tau)$, so the GCL-driven rate in Theorem 4.2 is, in closed form,

$$(6.2) \quad \gamma(\tau) = \frac{A\omega^2 |\sin(\omega\tau)|}{c - A\omega \cos(\omega\tau)}.$$

Figure 4(a) plots $\gamma(\tau)$ over one period; its maximum, $\max_{\tau} \gamma(\tau) \approx 5.07$, occurs where the grid deceleration is largest relative to the shrinking margin $a(\tau) = c - \dot{x}_M(\tau)$. Figure 4(b) plots the resulting envelope $\exp(\int_0^{\tau} \gamma(s) ds)$ (the $\rho(\tau)$ contribution from Assumption 4.1 is grid-dependent through $\lambda_{\min}(\hat{H})$ and is reported separately below, rather than folded into this plot, so that the purely motion-induced part of the bound can be seen on its own); by $\tau = 1$ (one full period of the prescribed motion) the bound has grown to $\exp(\int_0^1 \gamma ds) \approx 19.2$. This is a valid, rigorous, a priori upper bound: it proves the semi-discrete energy cannot blow up over the interval simulated. Consistent with Remark 4.4, it is also visibly conservative: the eigenvalue spectra in Figure 3, together with the observed L_2 accuracy in Figure 2, show no sign of the sharp, factor-of-nineteen energy amplification the bound permits, over the same time interval and grid motion. We regard this gap between the rigorous bound and

the observed behavior as expected — Young’s inequality and Grönwall’s lemma are not, in general, sharp — and as an indication that a tighter estimate (for instance, exploiting a more refined bound on the interpolation-coupling term (4.5) than the crude Assumption 4.1) is a natural target for future refinement, rather than evidence against the bound itself.

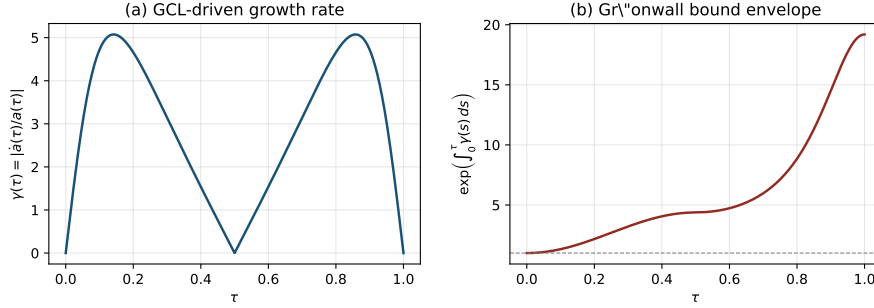


FIG. 4. (a) The GCL-driven rate $\gamma(\tau)$ of (6.2) over one period of the prescribed sinusoidal grid motion (6.1). (b) The corresponding Grönwall envelope $\exp(\int_0^\tau \gamma(s) ds)$, a rigorous a priori upper bound on the motion-induced part of the energy growth in Theorem 4.2.

Remark 6.1 (The interpolation constants Λ_L, Λ_M). For the 1-2-1 scheme, linear interpolation weights $(1 - \theta, \theta)$, $\theta \in [0, 1]$, give $\Lambda_L = \Lambda_M \leq 1$ exactly, with equality only in the degenerate limit $\theta \in \{0, 1\}$. For the cubic (4-point) Lagrange stencils used with the 2-4-2 and 3-6-3 schemes, $\Lambda = (\sum_i \ell_i(\theta)^2)^{1/2}$ is a fixed, computable function of the (grid-independent) offset θ of the interpolation point within the innermost sub-interval of the stencil (nodes at 0, 1, 2, 3, interpolation point at $1 + \theta$, $\theta \in [0, 1]$, consistent with the centered placement used to retain the interior order of accuracy). Evaluating $\Lambda(\theta)$ directly gives $\Lambda \leq 1$ on this interval as well, with the minimum $\Lambda \approx 0.80$ at $\theta = \frac{1}{2}$ and the maximum $\Lambda = 1$, attained (as in the linear case) only at the sub-interval endpoints $\theta \in \{0, 1\}$; this is consistent with the well-known smallness of Lebesgue constants for low-order Lagrange interpolation, and somewhat stronger than the generic bound one would expect a priori. We recommend that Λ_L, Λ_M be evaluated numerically for the exact interpolation stencils used, on the exact grids used, before Theorem 4.2 is invoked quantitatively in a given computation; the qualitative conclusion (energy remains finite, with an explicit, computable rate) does not depend on the particular value.

7. Grid dependence of the energy bound and possible improvements.

The estimate in Theorem 4.2 establishes boundedness on finite time intervals for each fixed spatial discretization. A stronger result would control a family of discretizations as the component grids are refined. Let h index such a family and let W_h denote the block-diagonal physical quadrature norm, whose blocks are $H_\kappa \text{diag}(x_{\xi, \kappa})$. The desired estimate, for homogeneous physical boundary data, is

$$(7.1) \quad \|\Phi_h(\tau)\|_{W_h}^2 \leq C_T \|\Phi_h(0)\|_{W_h}^2, \quad 0 \leq \tau \leq T,$$

with C_T independent of the component-grid spacings. Such a result would require specified mesh-regularity and overlap assumptions, a relative advection speed bounded away from zero, and constants uniform over all interpolation configurations encountered during the motion. It would be a semi-discrete stability estimate; convergence

and stability of a time-discrete implementation require additional consistency and time-integration arguments.

7.1. Origin of the grid dependence. The factor responsible for the loss of uniformity is $\rho = (\Lambda_L^2 + \Lambda_M^2)/\lambda_{\min}(\hat{H})$ in (4.6). For quasi-uniform component grids with comparable spacings, bounded mapping metrics, and a uniformly positive relative advection speed, $\lambda_{\min}(\hat{H})$ scales as h . Thus the estimate permits growth proportional to $\exp(O(T/h))$. This deterioration is a limitation of the estimate and does not, by itself, demonstrate instability of the discrete evolution. Related difficulties in obtaining bounds uniform in discretization parameters have also been identified for overset discontinuous Galerkin spectral element approximations [11].

The loss occurs when the interpolated donor value, which acts as boundary forcing for the receiver, is bounded by a donor volume norm. For a donor interpolation vector I and physical norm W_D , the sharp weighted Cauchy–Schwarz inequality is

$$(7.2) \quad |I^T \mathbf{z}|^2 \leq (I^T W_D^{-1} I) \|\mathbf{z}\|_{W_D}^2.$$

For a consistent interpolation stencil with a fixed number of points and bounded weights, $I^T W_D^{-1} I$ scales as h^{-1} on a quasi-uniform donor grid. Consequently, replacing the Euclidean bound in Assumption 4.1 with a weighted interpolation bound does not alone resolve the difficulty. A uniform estimate must use additional properties of the evolution or a coupling construction that balances the donor and receiver energy contributions.

7.2. A fixed physical energy for rigid translation. Before addressing the coupling, the energy can be simplified for the particular motion considered here. Define

$$(7.3) \quad W_M = H_M \Xi_x^{-1}, \quad E_M^{\text{phys}} = \mathbf{v}^T W_M \mathbf{v}, \quad q_M(\tau) = \mathbf{I}_L(\tau)^T \mathbf{u}(\tau).$$

Rigid translation makes W_M time-independent. Multiplication of (2.13) by $2\mathbf{v}^T W_M$, using (4.1), the SBP identity, and $\tau_M = 1$, gives

$$(7.4) \quad \frac{dE_M^{\text{phys}}}{d\tau} = a(\tau) [-v_0^2 - v_n^2 + 2v_0 q_M] = a(\tau) [q_M^2 - (v_0 - q_M)^2 - v_n^2].$$

There is no acceleration contribution in this identity. The term γ in Theorem 4.2 originates from differentiating the chosen weight $\hat{H}_M = W_M/a(\tau)$; it is not an unavoidable source of growth in the physical energy. The two weights are uniformly equivalent when a has positive lower and finite upper bounds independent of h . This change of energy therefore simplifies the analysis, but it leaves the donor-trace term $a q_M^2$ to be controlled. The use of physical energy balances is also central to the ALE framework of Lundquist, Malan, and Nordström [12].

7.3. Time-integrated donor-trace estimates. A possible route that preserves the current discretization exploits its one-way coupling, from the left grid to the middle grid and then to the right grid. Instead of applying (7.2) instantaneously, one could seek

$$(7.5) \quad \int_0^T |\mathbf{I}_L(\tau)^T \mathbf{u}(\tau)|^2 d\tau \leq C_T \|\mathbf{u}(0)\|_{W_L}^2$$

for homogeneous left-boundary data, with C_T independent of h . An analogous estimate for the middle-grid donor trace would include both its initial energy and its

time-integrated incoming data. Integrating balances such as (7.4) and applying these estimates successively would then control the energy of the three-grid system without the instantaneous inverse estimate.

The continuous advection equation suggests why this approach may be useful. Along a moving observation point $x_b(\tau)$, a characteristic originating at the initial line gives $u(x_b(\tau), \tau) = f(x_b(\tau) - c\tau)$. If $c - \dot{x}_b \geq a_* > 0$, a change of variables bounds the integral of the squared trace by $a_*^{-1} \|f\|_{L^2}^2$ over that part of the trajectory. Boundary-entering characteristics contribute a corresponding boundary-data term. A discrete counterpart is not established here: it must hold for all grid functions, including high-frequency modes, and uniformly through changes of donor stencil. Numerical dispersion can obstruct an argument based only on the continuous characteristic speed. Thus (7.5) is a proposed estimate to investigate, not a consequence of Assumption 4.1.

7.4. Overlap energy balances and compatible coupling. An alternative is to modify the energy and, if necessary, the interface treatment so that receiver inflow is balanced by a donor flux at the same physical location. The continuous overset analysis of Kopriva, Nordström, and Gassner [9] accounts explicitly for the overlap in the global energy. A discrete analogue requires integration by parts on donor subdomains ending at the overset interfaces, together with interpolation and SAT operators that make the combined interface quadratic form nonpositive. Sub-cell SBP operators provide a relevant construction for this purpose [6].

For translating interfaces, such a construction must additionally account for the motion of the overlap boundaries and the associated relative characteristic speeds. The overlap energy must remain uniformly equivalent to W_h as the grids move and are refined; vanishing weights or degenerating cut cells cannot be allowed to hide grid dependence in the norm-equivalence constants. If the resulting balance satisfies $\dot{E}_h \leq C(\tau)E_h$ with a uniformly integrable rate independent of h , then (7.1) follows. Extending the cited constructions to the present moving interfaces remains an additional task.

7.5. Uniformly equivalent energies for the coupled system. A further possibility is to retain the scheme and seek an energy matrix $P_h(\tau)$ with off-diagonal blocks representing interactions between component-grid solutions. Sufficient conditions are

$$(7.6) \quad mW_h \preceq P_h(\tau) \preceq MW_h, \quad \dot{P}_h + P_h M_h + M_h^T P_h \preceq C(\tau)P_h,$$

where $m, M > 0$ and $\int_0^T C(\tau) d\tau$ are bounded independently of h . These conditions imply (7.1) with $C_T = (M/m) \exp(\int_0^T C(\tau) d\tau)$. Existence of a Lyapunov matrix at each fixed resolution is insufficient unless these uniform bounds, including the contribution from $\dot{P}_h$, are verified.

As a diagnostic, the physical-norm amplification of the full non-autonomous propagator $U_h(\tau, 0)$ can be examined through $\|W_h^{1/2} U_h(\tau, 0) W_h^{-1/2}\|_2$ under refinement. This measures transient growth that frozen-time eigenvalues do not resolve, although finite numerical tests cannot establish a uniform bound. For the present scheme, the fixed physical energy and successive donor-trace estimates offer a direct starting point. If uniform trace control cannot be obtained, a coupling based on an overlap energy balance is a natural alternative. Neither increasing the interpolation order nor imposing geometric conservation alone supplies the missing interface estimate.

8. Conclusions. We have shown, for the model problem of weak SBP-SAT overset coupling of the scalar advection equation on a rigidly translating grid, that the

standard static-grid stability argument — verifying that the semi-discrete system matrix is Hurwitz — generalizes to a *necessary but not sufficient* frozen-time condition once the grid moves, and we have identified precisely the term, $\Phi^T(d\hat{H}/d\tau)\Phi$, that the frozen-time argument omits. Contrary to a natural first intuition, this term is not rendered negligible by grid refinement: its point-wise $O(h)$ scaling is exactly offset, in aggregate, by the $O(1/h)$ number of grid points it acts over. We have proved that, for rigid translation, this term collapses through a discrete geometric-conservation-law identity to a single explicit scalar rate, and that the resulting non-autonomous energy inequality closes via Grönwall’s lemma under a mild, verifiable assumption on the interpolation operators, yielding a rigorous, a priori, computable bound on the growth of the semi-discrete energy. This bound applies at fixed spatial resolution; Section 7 identifies the donor-trace estimate as the source of its grid dependence and discusses possible improvements. The acceleration contribution can be removed by using a fixed physical energy for rigid translation, but uniform control of the overset coupling remains to be established. We have deliberately kept the scope of this result narrow — one space dimension, a scalar equation, rigid translation — because each of these restrictions is used essentially in the proof, and we have outlined, as a scoped remark rather than a theorem, the substantially larger body of structure (entropy variables, entropy-stable SATs, a discrete GCL for the nonlinear metric Jacobian) that extending the result to rotating or deforming grids and to the nonlinear compressible Euler and Navier–Stokes equations would require.

Appendix A. Notation. Throughout, $\kappa \in \{L, M, R\}$ indexes the left, middle, and right grids; H_κ is the diagonal SBP norm matrix on grid κ ; $\mathcal{X}_\kappa$ is the diagonal matrix of local wave-speed-relative- to-grid-speed metric factors; $\hat{H}_\kappa = H_\kappa \mathcal{X}_\kappa^{-1}$; e_0^κ selects the upwind (left) boundary node of grid κ ; τ_κ is the SAT penalty parameter; $\mathbf{I}_L(\tau)$, $\mathbf{I}_M(\tau)$ are the (generally time-dependent) interpolation weight vectors from the left grid to the middle grid, and from the middle grid to the right grid, respectively; a dot or subscript τ denotes $d/d\tau$; $\preceq$ denotes the negative semi-definite matrix ordering.

REFERENCES

- [1] P. J. ANTSAKLIS AND A. N. MICHEL, *A Linear Systems Primer*, Birkhäuser Basel, 1st ed., 2007.
- [2] D. J. BODONY, G. ZAGARIS, A. REICHERT, AND Q. ZHANG, *Provably stable overset grid methods for computational aeroacoustics*, Journal of Sound and Vibration, 330 (2011), pp. 4161–4179.
- [3] G. CHESSHIRE AND W. HENSHAW, *A scheme for conservative interpolation on overlapping grids*, SIAM Journal on Scientific Computing, 15 (1994), pp. 819–845.
- [4] G. CHESSHIRE AND W. D. HENSHAW, *Composite overlapping meshes for the solution of partial differential equations*, Journal of Computational Physics, 90 (1990), pp. 1–64.
- [5] T. C. FISHER AND M. H. CARPENTER, *High-order entropy stable finite difference schemes for nonlinear conservation laws: Finite domains*, Journal of Computational Physics, 252 (2013), pp. 518–557.
- [6] J. GLAUBITZ, J. LAMPERT, A. R. WINTERS, AND J. NORDSTRÖM, *Towards provable energy-stable overset grid methods using sub-cell summation-by-parts operators*, 2025, <https://arxiv.org/abs/2509.21442>. arXiv preprint arXiv:2509.21442.
- [7] B. GUSTAFSSON, *The convergence rate for difference approximations to mixed initial boundary value problems*, Mathematics of Computation, 29 (1975), pp. 396–406.
- [8] B. GUSTAFSSON, *High order difference methods for time dependent PDE*, vol. 38, Springer Science & Business Media, 2007.
- [9] D. A. KOPRIVA, J. NORDSTRÖM, AND G. J. GASSNER, *On the theoretical foundation of overset grid methods for hyperbolic problems: Well-posedness and conservation*, Journal of Computational Physics, 448 (2022), p. 110732, <https://doi.org/10.1016/j.jcp.2021.110732>.

- [10] D. A. KOPRIVA, A. R. WINTERS, M. BOHM, AND G. J. GASSNER, *A provably stable discontinuous Galerkin spectral element approximation for moving hexahedral meshes*, *Computers & Fluids*, 139 (2016), pp. 148–160, <https://doi.org/10.1016/j.compfluid.2016.05.023>.
- [11] D. A. KOPRIVA, A. R. WINTERS, AND J. NORDSTRÖM, *Energy bounds for discontinuous Galerkin spectral element approximations of well-posed overset grid problems for hyperbolic systems*, *Journal of Computational Physics*, 520 (2025), p. 113508, <https://doi.org/10.1016/j.jcp.2024.113508>.
- [12] T. LUNDQUIST, A. G. MALAN, AND J. NORDSTRÖM, *A method-of-lines framework for energy stable arbitrary Lagrangian–Eulerian methods*, *SIAM Journal on Numerical Analysis*, 61 (2023), pp. 2327–2351, <https://doi.org/10.1137/22M1514945>.
- [13] N. C. PREWITT, D. M. BELK, AND W. SHYY, *Parallel computing of overset grids for aerodynamic problems with moving objects*, *Progress in Aerospace Sciences*, 36 (2000), pp. 117–172.
- [14] A. REICHERT, M. T. HEATH, AND D. J. BODONY, *Energy stable numerical methods for hyperbolic partial differential equations using overlapping domain decomposition*, *Journal of Computational Physics*, 231 (2012), pp. 5243–5265.
- [15] W. J. RUGH, *Linear system theory*, Prentice-Hall, Inc., 1996.
- [16] G. SCHNÜCKE, N. KRAIS, T. BOLEMAN, AND G. J. GASSNER, *Entropy stable discontinuous Galerkin schemes on moving meshes for hyperbolic conservation laws*, *Journal of Scientific Computing*, 82 (2020), p. 69, <https://doi.org/10.1007/s10915-020-01171-7>.
- [17] N. SHARAN, C. PANTANO, AND D. J. BODONY, *Energy stable overset grid methods for hyperbolic problems*, in *7th AIAA Theoretical Fluid Mechanics Conference*, 2014, p. 2924.
- [18] N. SHARAN, C. PANTANO, AND D. J. BODONY, *Time-stable overset grid method for hyperbolic problems using summation-by-parts operators*, *Journal of Computational Physics*, 361 (2018), pp. 199–230.
- [19] S. E. SHERER AND J. N. SCOTT, *High-order compact finite-difference methods on general overset grids*, *Journal of Computational Physics*, 210 (2005), pp. 459–496.
- [20] M. SVÄRD AND J. NORDSTRÖM, *Review of summation-by-parts schemes for initial-boundary-value problems*, *Journal of Computational Physics*, 268 (2014), pp. 17–38, <https://doi.org/10.1016/j.jcp.2014.02.031>.
- [21] E. TADMOR, *Entropy stability theory for difference approximations of nonlinear conservation laws and related time-dependent problems*, *Acta Numerica*, 12 (2003), pp. 451–512.
- [22] M. R. VISBAL AND D. V. GAITONDE, *On the use of higher-order finite-difference schemes on curvilinear and deforming meshes*, *Journal of Computational Physics*, 181 (2002), pp. 155–185.
- [23] N. K. YAMALEEV, D. C. DEL REY FERNÁNDEZ, J. LOU, AND M. H. CARPENTER, *Entropy stable spectral collocation schemes for the 3-D Navier–Stokes equations on dynamic unstructured grids*, *Journal of Computational Physics*, 399 (2019), p. 108897, <https://doi.org/10.1016/j.jcp.2019.108897>.